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Elements, Compounds & Mixtures Worksheet

Part 1: Read the following information on elements, compounds and mixtures. Fill in the blanks where necessary.

Elements:

- A pure substance containing only one kind of _____.
- An element is always uniform all the way through (homogeneous).
- An element _____ be separated into simpler materials (except during nuclear reactions).
- Over 100 existing elements are listed and classified on the _____.

Compounds:

- A pure substance containing two or more kinds of _____.
- The atoms are _____ combined in some way. Often times (but not always) they come together to form groups of atoms called molecules.
- A compound is always homogeneous (uniform).
- Compounds _____ be separated by physical means. Separating a compound requires a chemical reaction.
- The properties of a compound are usually different than the properties of the elements it contains.

Mixtures:

- Two or more _____ or _____ NOT chemically combined.
- No reaction between substances.
- Mixtures can be uniform (called _____) and are known as solutions.
- Mixtures can also be non-uniform (called _____).
- Mixtures can be separated into their components by chemical or physical means.
- The properties of a mixture are similar to the properties of its components.

Part 2: Classify each of the following as elements (E), compounds (C) or Mixtures (M). Write the letter X if it is none of these.

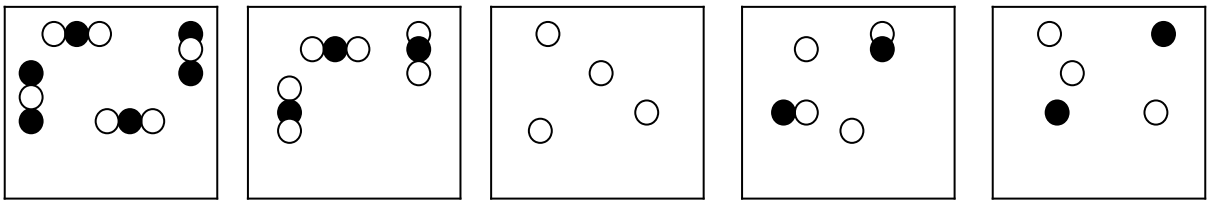
___Diamond (C) ___Sugar ($C_6H_{12}O_6$) ___Milk ___Iron (Fe)
___Air ___Sulfuric Acid (H_2SO_4) ___Gasoline ___Electricity
___Krypton (K) ___Bismuth (Bi) ___Uranium (U) ___Popcorn
___Water (H_2O) ___Alcohol (CH_3OH) ___Pail of Garbage ___A dog
___Ammonia (NH_3) ___Salt (NaCl) ___Energy ___Gold (Au)

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___Wood ___Bronze ___Ink ___Pizza
___Dry Ice (CO_2) ___Baking Soda ($NaHCO_3$) ___Titanium (Ti)
___Concrete

Part 3: Match each diagram with its correct description. Diagrams will be used once.



A **B** **C** **D** **E**

___1. Pure Element – only one type of atom present.

___2. Mixture of two elements – two types of uncombined atoms present.

___3. Pure compound – only one type of compound present.

___4. Mixture of two compounds – two types of compounds present.

___5. Mixture of a compound and an element.

Part 4: Column A lists a substance. In Column B, list whether the substance is an element (E), a compound (C), a Heterogeneous Mixture (HM), or a Solution (S). (Remember a solution is a homogeneous mixture.) In Column C, list TWO physical properties of the substance.

Column A	Column B	Column C
1. Summer Sausage		
2. Steam		
3. Salt Water		
4. Pencil lead (Pb)		
5. Dirt		
6. Pepsi		
7. Silver (Ag)		
8. Toothpaste (Na_2HPO_4)		
9. A burrito		
10. Italian Dressing		
11. Chicken Soup		

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12. Lemonade		

Elements, Compounds & Mixtures Worksheet

Part 1: Read the following information on elements, compounds and mixtures. Fill in the blanks where necessary.

Elements:

- A pure substance containing only one kind of __**atom**____.
- An element is always uniform all the way through (homogeneous).
- An element __**cannot**___ be separated into simpler materials (except during nuclear reactions).
- Over 100 existing elements are listed and classified on the _**Periodic Table**_.

Compounds:

- A pure substance containing two or more kinds of __**atoms**__.
- The atoms are ___**chemically**___ combined in some way. Often times (but not always) they come together to form groups of atoms called molecules.
- A compound is always homogeneous (uniform).
- Compounds ___**cannot**___ be separated by physical means. Separating a compound requires a chemical reaction.
- The properties of a compound are usually different than the properties of the elements it contains.

Mixtures:

- Two or more __**elements**___ or ____**compounds**__ NOT chemically combined.
- No reaction between substances.
- Mixtures can be uniform (called __**homogeneous**___) and are known as solutions.
- Mixtures can also be non-uniform (called ____**heterogeneous**____).
- Mixtures can be separated into their components by chemical or physical means.
- The properties of a mixture are similar to the properties of its components.

Part 2: Classify each of the following as elements (E), compounds (C) or Mixtures (M). Write the letter X if it is none of these.

_**E**_Diamond (C) _**C**_Sugar ($C_6H_{12}O_6$) _**M**_Milk

_**E**_Iron (Fe)

_**M**_Air _**C**_Sulfuric Acid (H_2SO_4) _**M**_Gasoline

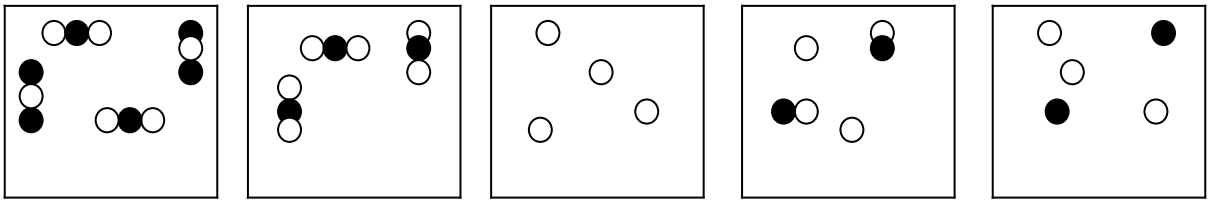
_**X**_Electricity

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_**E**_Krypton (K) _**E**_Bismuth (Bi) _**E**_Uranium (U) _**M**_Popcorn
_**C**_Water (H_2O) _**C**_Alcohol (CH_3OH) _**M**_Pail of Garbage _**M**_A dog
_**C**_Ammonia (NH_3) _**C**_Salt (NaCl) _**X**_Energy _**E**_Gold (Au)
_**M**_Wood _**M**_Bronze _**M**_Ink _**M**_Pizza
_**C**_Dry Ice (CO_2) _**C**_Baking Soda ($NaHCO_3$) _**E**_Titanium (Ti)
_**M**_Concrete

Part 3: Match each diagram with its correct description. Diagrams will be used once.



A **B** **C** **D** **E**

_**C**_1. Pure Element – only one type of atom present.

_**E**_2. Mixture of two elements – two types of uncombined atoms present.

_**B**_3. Pure compound – only one type of compound present.

_**A**_4. Mixture of two compounds – two types of compounds present.

_**D**_5. Mixture of a compound and an element.

Part 4: Column A lists a substance. In Column B, list whether the substance is an element (E), a compound (C), a Heterogeneous Mixture (HM), or a Solution (S). (Remember a solution is a homogeneous mixture.) In Column C, list TWO physical properties of the substance.

Column A	Column B	Column C
1. Summer Sausage	**HM**	**Chunky, Brown**
2. Steam	**C**	**Gas, Hot**
3. Salt Water	**S**	**Liquid, Clear**
4. Pencil lead (Pb)	**E**	**Grey, Solid**
5. Dirt	**HM**	**Brown, Solid**
6. Pepsi	**HM**	**Brown, Liquid**
7. Silver (Ag)	**E**	**Silver, Solid**
8. Toothpaste (Na_2HPO_4)	**C**	**White, Thick**
9. A burrito	**HM**	**Multi-colored, Solid**
10. Italian Dressing	**HM**	**Liquid, Greasy**

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11. Chicken Soup	**HM**	**Liquid/Solid, Brown**
12. Lemonade	**S**	**Yellow, Liquid**